

Delivering a 331% Performance Leap in Medical AI

Build a specialized Drug-Drug Interaction (DDI) inference engine capable of deciphering complex pharmacology.



331.68%

Performance Increase vs Base Model

Medical Accuracy Score: 8.245 (Fine-Tuned) vs 1.91 (Base).



Engineering Enterprise-Scale Generative AI



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Generative AI Architect

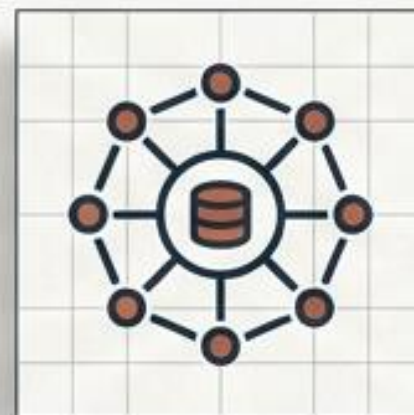
Architect's Note

"Applying distributed systems principles from enterprise mobile engineering to deploy production-grade, 3-tier GenAI applications with zero-idle-cost architectures."



**15+ Years
High-Availability
Mobile Architecture**
(CVS Health, McDonald's)

**Production-Grade
Generative AI &
Multi-Agent
Orchestrator**



Multi-Agent Orchestration

Google Agent Engine, ADK 1.20, LangGraph.



LLMOps & Observability

Telemetry Gravity architecture, Arize Phoenix, GCP Cloud Trace.



Model Engineering

Domain-specific PEFT (LoRA, Unsloth), GGUF Quantization, Custom RAG pipelines.

The End-to-End Medical AI MLOps Pipeline



Phase 1: Data Wrangling & Curation
PyTDC, Vocab Mapping



Phase 2: Fine-Tuning & Resource Optimization
RTX 4090, PEFT



Phase 3: Evaluation & Model Conversion
LLM-as-a-Judge



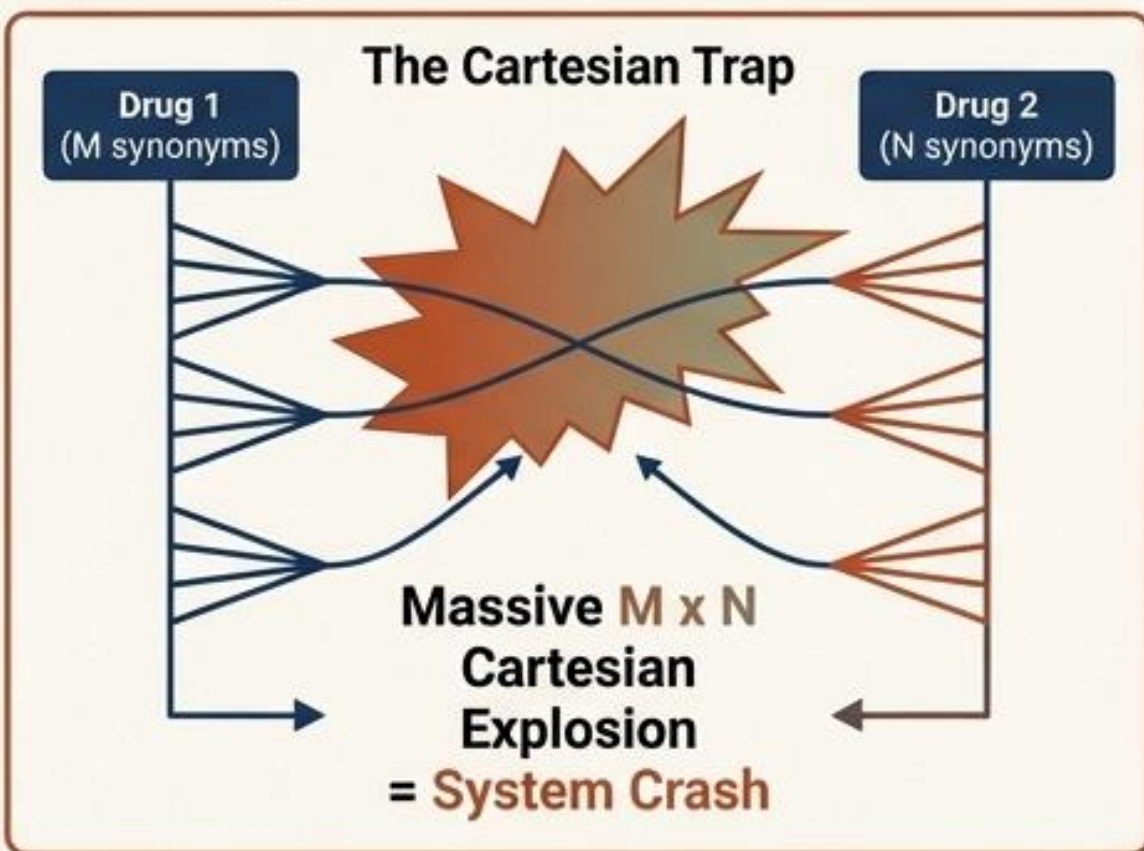
Phase 4: Secure 3-Tier Cloud Deployment
Google Cloud Platform

Wrangling 300k Records to Prevent Garbage In, Garbage Out

The Challenge

Raw PyTDC dataset contained **missing vocabularies** and isolated numerical IDs (86 distinct interactions).

Feeding all drug synonyms into a single pipe-delimited prompt confused the model, resulting in poor, single-word outputs.



The Resolution

Built a unique **local master vocabulary** to map missing drugs.

Capped data augmentation to exactly three random synonym pairs per record.

Tripled the training set to ~300k clean, highly generalizable conversational pairs mapped to native LLM chat templates.

Architect's Note

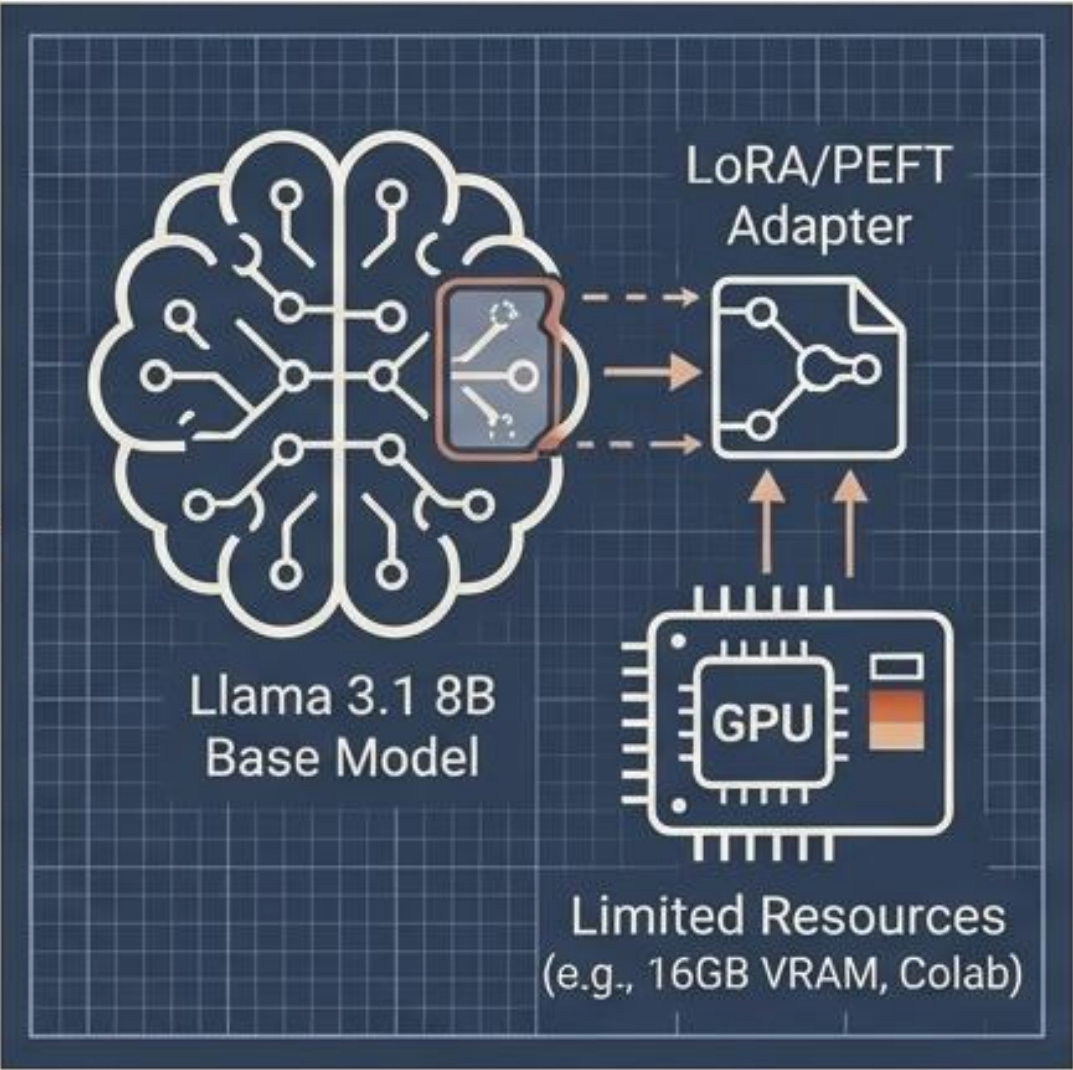
If you speak Spanish, I have to speak Spanish. Translating augmented pairs into the exact chat template the LLM natively expects is non-negotiable for fine-tuning.

Overcoming Hardware Constraints for Llama 3.1 8B

The Challenge



Standard consumer GPUs (16GB VRAM) and cloud notebooks repeatedly crashed under the weight of the 8B parameter model and subsequent storage limits.



The Resolution



Shifted to a rented RTX 4090 instance to bypass memory bottlenecks and storage limits.

Infrastructure Specs

GPU	VRAM	Storage
RTX 4090	40GB (Allowed batch size scaling)	80GB Container (Crucial for saving multiple GGUF formats)

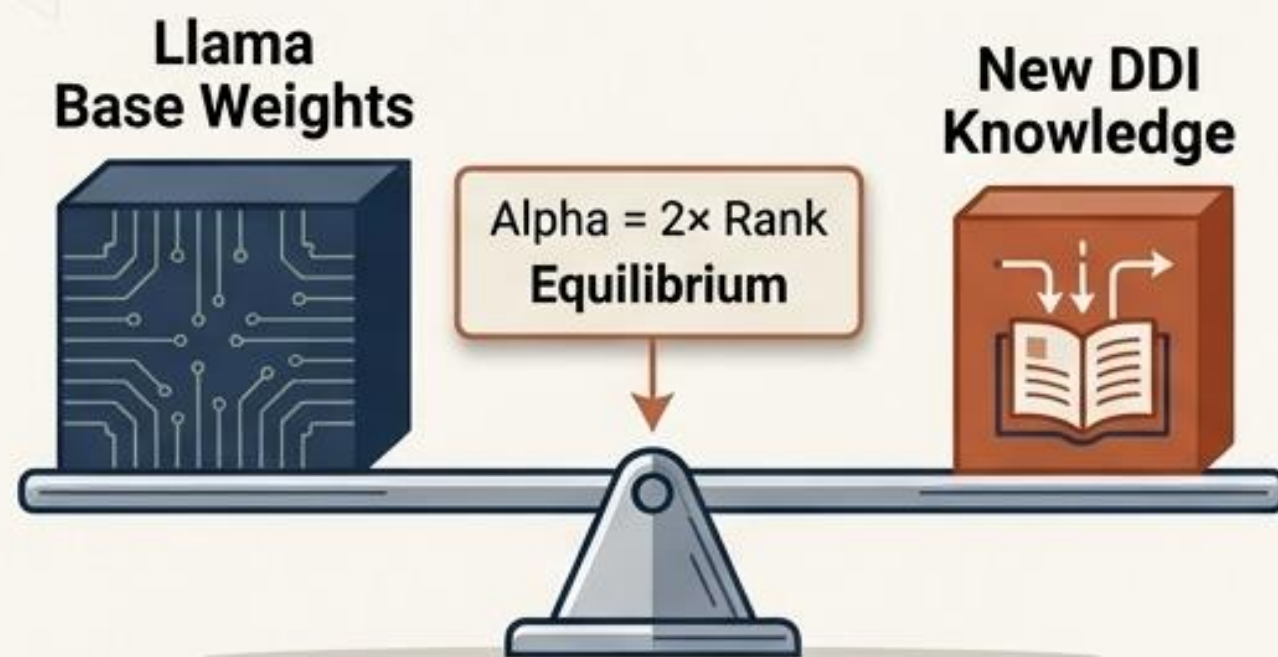


Architect's Note

Parameter-Efficient Fine-Tuning (PEFT) is like ordering pre-prepped ingredients instead of cooking from scratch. By using Unsloth and LoRA adapters with a Rank of 16, we only trained 0.52% of the total model parameters.

The Math of Generalization and Alpha Balancing

Preventing Catastrophic Forgetting

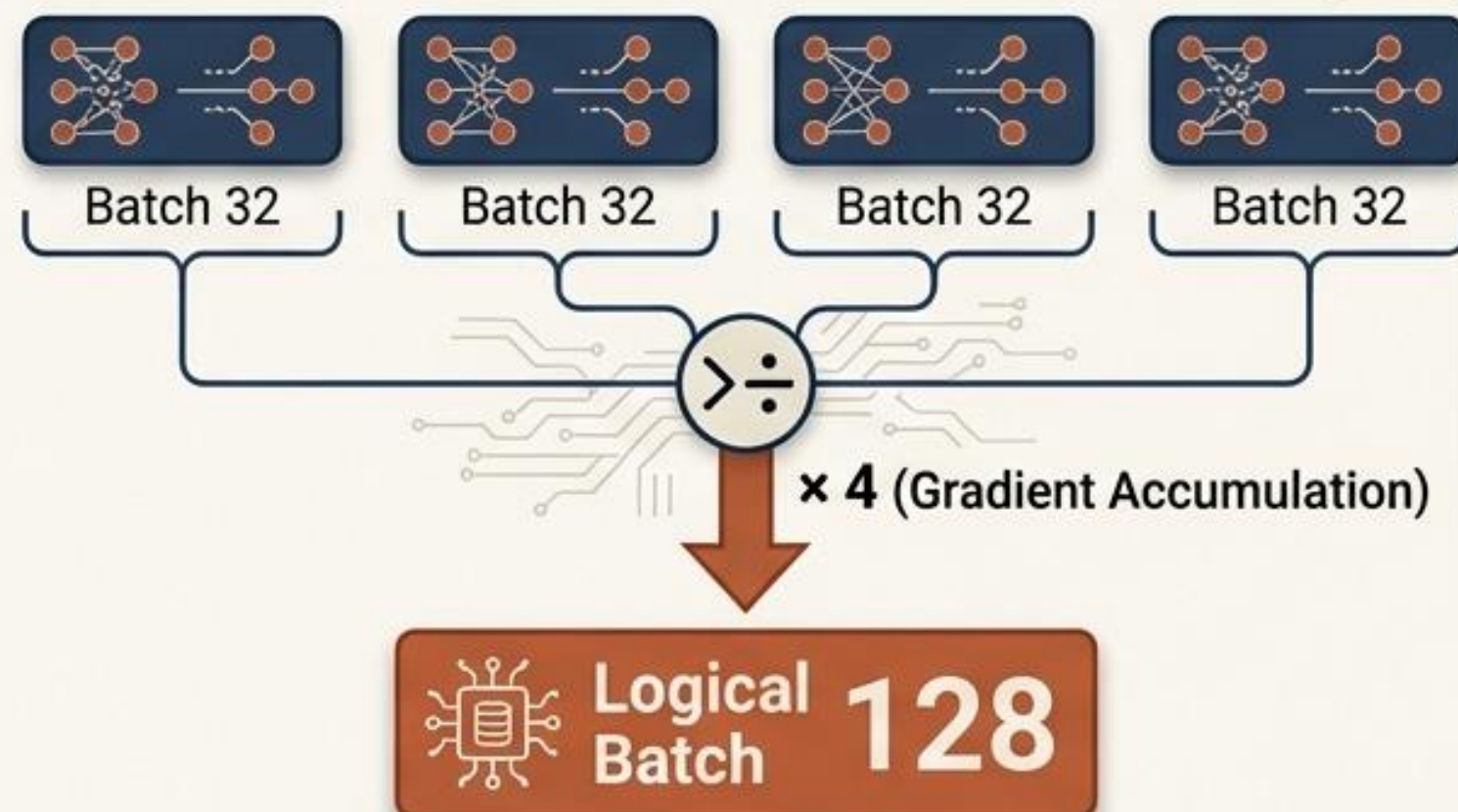


Concept: Setting the LoRA 'Alpha' parameter incorrectly causes the model to become hypnotized by new data, overwriting base knowledge.

Resolution: Balanced Alpha at strictly 1x or 2x the Rank to maintain equilibrium.



Logical Batching



Impact: Forces the model to read 128 records simultaneously before updating weights, ensuring broad generalization rather than single-record bias.



Optimization Stack:



Adam 8-bit Optimizer



Linear Learning Rate Schedule



21-Minute Training Time

Engineering an Unbiased LLM-as-a-Judge Evaluation

⚠ Challenge

Traditional metrics (F1, Accuracy) fail on generative outputs. Unsloth's native GGUF exporter failed because the base model was pre-loaded in a 4-bit quantized state.

🔧 Format Resolution

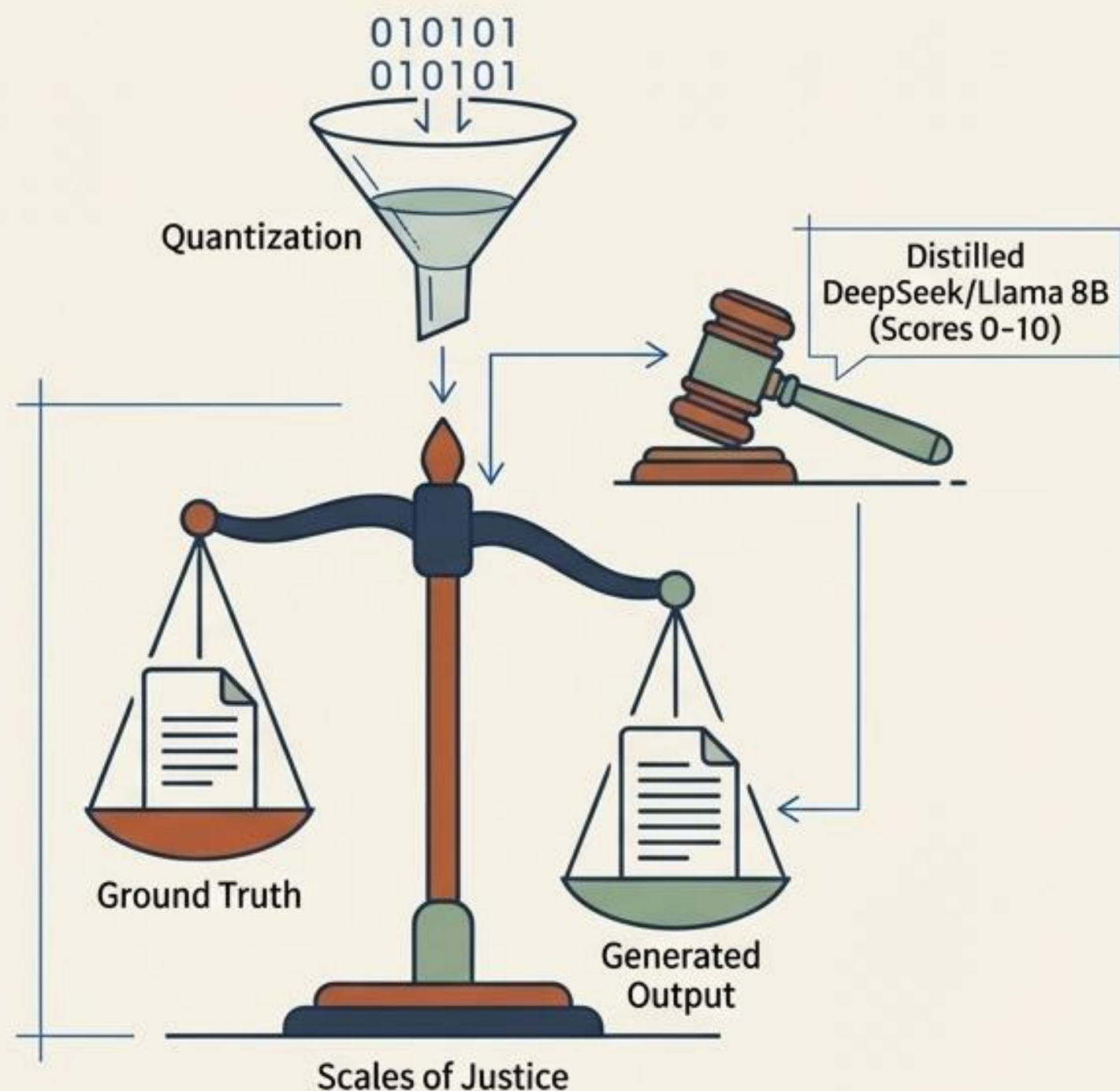
Utilized llama.cpp to bypass architectural conflicts, successfully converting the models to Q8 and Q4_K_M GGUF formats.

✅ Evaluation Resolution

Engineered a multi-persona 'LLM-as-a-Judge' pipeline.

Architect's Note

Tested against a strictly unseen validation dataset to ensure zero data leakage. If the model didn't provide an adequate conversational response, the judge scored it a zero.



Quantifying the 331% Inference Performance Leap

Llama 3.1 8B Base Model

- Medical Accuracy Score: 1.91 / 10
- Output Quality: Hallucinated drug existence; provided poor, single-word responses.



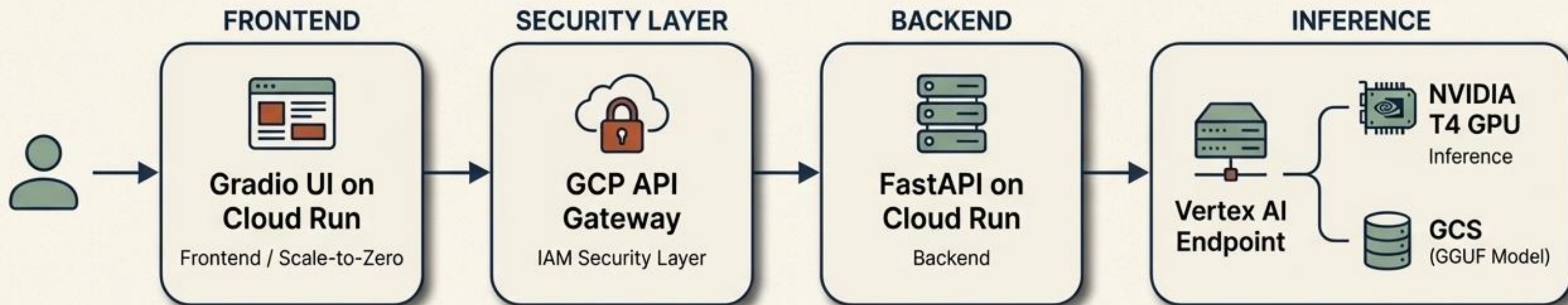
Fine-Tuned DDI Model

- Medical Accuracy Score: 8.245 / 10
- Output Quality: Accurate, conversational interactions; highly specialized recognition.

Architect's Note:

For local macOS demoing, 32GB system RAM couldn't support two 16GB FP16 models. I used a Kaggle instance to manually convert the base model to an 8-bit Q8 GGUF, enabling dual-loading for direct comparison.

Navigating the Deployment Gauntlet on Google Cloud



The Gauntlet



504 Gateway Timeouts

Vertex AI cold starts and GCS download latency caused timeouts. Resolved by optimizing container startup sequences and GGUF loading logic.



403 Forbidden Errors

Inter-service authentication failures. Resolved via a secure IAM Authentication Bridge with strict Invoker roles.

Advancing Enterprise Multi-Agent Architecture



Architect's Note (Final Takeaway)

From single-model domain fine-tuning to distributed multi-agent RAG pipelines, true enterprise AI requires a unified approach to telemetry, reliability, and scale-to-zero infrastructure.